

CSE 400: Fundamentals of Probability in Computing

Lecture 23 Scribe: Stationary Distributions of Markov Chains

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Lecture Purpose

This scribe is intended as authoritative reference material for exam preparation. It reconstructs the logical flow of Lecture 23, focusing on the long-term behavior of discrete-time Markov chains and the calculation of stationary distributions.

1 Definitions and Notation

1.1 Stationary Distribution

A probability distribution $\pi = [\pi_1, \pi_2, \dots, \pi_n]$ is called a **stationary distribution** for a Markov chain with transition probability matrix P if it satisfies the following conditions:

1. **Balance Equation:**

$$\pi P = \pi$$

2. **Normalization Condition:**

$$\sum_{i=1}^n \pi_i = 1$$

3. **Probability Constraint:**

$$\pi_i \geq 0 \quad \text{for all } i$$

2 Existence and Uniqueness

2.1 Assumptions and Conditions

According to the lecture slides, the stationary distribution exists and is unique if the Markov chain satisfies the following properties:

- **Irreducibility:** It is possible to reach any state from any other state in a finite number of steps.
- **Aperiodicity:** The states do not have a periodic recurrence pattern.
- **Finite State Space:** The number of states is finite.

3 The Stationary Theorem

If a Markov chain is irreducible and aperiodic, then:

1. A unique stationary distribution π exists.
2. For any initial state i , the n -step transition probabilities satisfy:

$$\lim_{n \rightarrow \infty} P_{ij}^n = \pi_j$$

4 Step-by-Step Reasoning: Solving for π

To find the stationary distribution, we follow these explicit steps derived from the lecture logic:

1. Write down the matrix equation $\pi = \pi P$.
2. Expand this into a system of n linear equations.
3. Note that the equations in the system are linearly dependent (the rank is $n - 1$).
4. Replace one of the redundant equations with the normalization condition $\sum \pi_i = 1$.
5. Solve the resulting system of linear equations using substitution or Gaussian elimination.

5 Worked Example: 2-State Markov Chain

Problem: Consider a 2-state Markov chain with states $\{0, 1\}$ and the following transition matrix P :

$$P = \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix}$$

Find the stationary distribution $\pi = [\pi_0, \pi_1]$.

Step-by-Step Solution:

1. **Set up the system** $\pi P = \pi$:

$$[\pi_0, \pi_1] \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix} = [\pi_0, \pi_1]$$

2. **Expand the equations:**

- $\pi_0(1 - \alpha) + \pi_1(\beta) = \pi_0 \implies -\alpha\pi_0 + \beta\pi_1 = 0 \implies \alpha\pi_0 = \beta\pi_1$
- $\pi_0(\alpha) + \pi_1(1 - \beta) = \pi_1 \implies \alpha\pi_0 - \beta\pi_1 = 0$ (This is the redundant equation)

3. **Apply Normalization:** $\pi_0 + \pi_1 = 1 \implies \pi_1 = 1 - \pi_0$.

4. **Substitute and Solve:**

$$\alpha\pi_0 = \beta(1 - \pi_0) \implies \alpha\pi_0 = \beta - \beta\pi_0$$

$$\pi_0(\alpha + \beta) = \beta \implies \pi_0 = \frac{\beta}{\alpha + \beta}$$

5. **Find π_1 :**

$$\pi_1 = 1 - \frac{\beta}{\alpha + \beta} = \frac{\alpha}{\alpha + \beta}$$

Result: The stationary distribution is $\pi = \left[\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right]$.

End of Lecture Scribe — Prepared strictly from Lecture Slides for exam revision.